Pawan Surya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.43
Narayana Junior College	Class XII (IPE)	2023	96.90%
KENNEDY HIGH SCHOOL	Class X (CBSE)	2021	85.80%

EXPERIENCE

Research Intern, Thomson Reuters Labs -- May 2026 - Present

Agentic Automation for Enterprise Workflows | Python, FastAPI, Claude Agent SDK, AWS, Linux

- Designed an **automated pipeline** that ingests large, irregular r**eal-world datasets**, detects the structure needed to normalize them to a target schema, and drives a **regulated downstream system** end to end
- Architected **human-in-the-loop** authorization for automated system writes using single-use, payload-bound approval tokens, so no duplicate or **unreviewed** write **reaches production data**
- Built **real-time observability** across a multi-service backend on **Braintrust**, tracing execution end to end and adding idempotent writes that close silent-failure paths
- Delivered incremental changes into a **live production** codebase through **PR review**, automated tests, and **Azure DevOps** backlog tracking alongside a **distributed engineering team**

Hierarchical Reasoning Model (HRM) Pre-training Research | PyTorch, SageMaker, MLflow, LightEval

- Built a 49.55M-parameter **transformer baseline** matching HRM's exact Attention, SwiGLU, RMSNorm, and RoPE stack and embeddings, isolating whether hierarchical recurrence **improves performance** independent of parameter count
- Adapted **HRM's masking** and loss formulation from fixed-answer puzzles to **open-ended language modeling** using **Prefix-LM** attention masks, answer-only loss, and EMA averaging
- Analyzed benchmark results to **identify a gap** where prefix-LM masks were silently skipped during evaluation scoring, then rebuilt tiered evaluation suites across **MMLU, GSM8K**, and five other benchmarks to confirm the fix.
- Diagnosed **NaN divergence** down to batch-size instability and resolved CUDA out-of-memory failures on multi-GPU Linux compute using **PyTorch's expandable-segments allocator**

PROJECTS

Temporal Modeling of AMD Onset from Longitudinal EHR [Link] *(Team Project)*

- Built a **multi-horizon risk model** predicting AMD onset across six clinical horizons (next visit to five years) from **irregular ICD-10 visit** sequences of 5,578 patients and 14,335 age- and sex-matched controls
- Initialized code embeddings from **PCA-projected ClinicalBERT** encodings of diagnosis text, raising T1 AUROC from **0.694** to **0.794** in ablation
- Benchmarked **LSTM, Mamba** (selective state-space), and transformer **(BEHRT)** sequence encoders with age-aware positional encoding BEHRT led every horizon, reaching 0.794 AUROC at T1 against LSTM's 0.743.
- Extended the model to **supervised-contrastive** and **DeepHit** discrete-time survival formulations, then used **attention-weight analysis** to surface clinically interpretable precursor codes.

Emotion-Cause Identification [Link] *(Team Project)*

- Built a **dual-head architecture** on a shared dialogue-aware encoder (a global emotion classifier on [CLS], a binary pairwise cause classifier) over a 2,398-dialogue dataset, benchmarking **BERT, RoBERTa,** and **DeBERTa-v3** backbones
- RoBERTa led both heads, reaching **61.1% accuracy / 91.2% top-3** for emotion and **85.1% accuracy** for cause extraction
- Designed an emotion-cause-enhanced **prompting scheme** feeding the classification stage directly into **FLAN-T5** summarization, improving BLEU-4, ROUGE-1, and BERTScore over both baselines
- Built an interactive **Streamlit demo** with live backbone switching and Mermaid.js causal-graph visualization
- Resolved an 8GB VRAM ceiling by **dynamically swapping** FLAN-T5-large between CPU and GPU, cutting idle memory use from **6GB+ to near zero**

Perceptual Trajectory Straightening in Video Diffusion [Link] *(Team Project)*

- Attacked **temporal flicker in diffusion video** generation by treating the latent frame sequence as a trajectory and regularizing its geometry, rather than filtering frames locally
- Implemented and ablated five **Retinal Latent Constraint** phases, diagnosing that positional corrections cannot be applied to velocity-like noise predictions
- Built **polynomial trajectory regression** via Vandermonde least-squares projection in perceptual and encoder latent space, evaluated with **curvature-based smoothness metrics**
- Improved temporal consistency across SDXL, SD 1.5, and PixArt: SSIM up **0.5-0.7%,** pixel-wise temporal difference down **18-21%**

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant** for the course of **Probability and Random Variables.** ***(Jan 2025 - April 2025)***
- **Elected as Mess Secretary for the tenure 25-26 in Student Gymkhana.**

EXTRACURRICULAR

Part of the campus **NSO football** team.